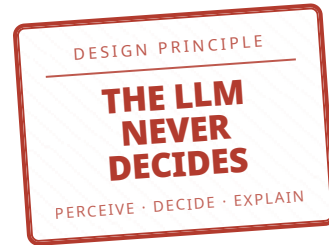


HACKATHON ET AI 2026	PROBLEM #4 · EPC Intelligence	TEAM HashForge	DOCUMENT Detailed Project Report
AUTHOR Awnikant Ajay	INSTITUTION IIIT Kalyani	REVISION R2 · 2026-07-22	CLASSIFICATION Prototype · Offline-first

00 AI INTELLIGENCE PLATFORM

SiteMind.



An AI intelligence layer that catches data-centre spec deviations before they reach site, and proves every call it makes.

Live prototype → sitemind.awni.in

SiteMind is an automated senior-reviewer for data-centre construction. It reads a Design Basis Report or vendor submittal, pulls out each engineering parameter with the exact sentence it came from, and checks it against the real Indian code clause that governs it. In seconds it returns cited non-conformances, a project copilot, schedule-risk forecasts, supply-chain visibility, and commissioning quality checks.

Its defining choice: **the pass-or-fail decision is made by deterministic Python against a cited clause, never by the language model.** The model only reads the document and writes the report; it is handed the clause it must cite and given no authority to decide. That is what makes SiteMind auditable instead of a "trust me" black box, and every number in this report is measured, not asserted.

100% Compliance-decision accuracy (macro-F1 1.0) vs 58.5% naive baseline, n=41	0% Hallucinated citations 32 of 32 cited clauses resolve to real text	21/21 Automated eval scripts passing reported per pillar, never blended	6,206 Verbatim-verified standard passages indexed across 17 real Indian codes
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Every number here is produced by a re-runnable evaluation script (`backend/eval/`), not asserted. The standards, citations, and checking logic are **real**; the pre-loaded project is **synthetic, modelled on public Indian data-centre tenders**. We disclose that openly, not as a footnote, because a tool built on honesty has to earn it.

01 THE PROBLEM

The intelligence to connect project data never gets built

India's data-centre capacity is set to grow from about 900 MW in 2024 to over 2,700 MW by 2027, more than \$15 billion of capital. A single hyperscale build carries **15,000 to 40,000 equipment line items**, up to **200 concurrent trade contractors**, and thousands of commissioning tests, at zero tolerance for anything that would dent the future uptime SLA.

WHAT ACTUALLY GOES WRONG

A 2024 Turner & Townsend survey found **67% of APAC data-centre EPC projects overran schedule by more than 10%**, led by procurement misalignment and commissioning failures.

The root cause is **information fragmentation**. Specifications, vendor submittals, test records, RFI logs, and change orders live in disconnected systems. A spec deviation is cheap to fix on paper and ruinous to fix on site, but nobody catches it in time, because the intelligence that would connect these documents is never built.

THE CHALLENGE

Build an AI intelligence layer over project data that enables **proactive schedule management**, **automated compliance and quality checking**, and **real-time commissioning support** across the full project lifecycle. The brief asks for impact measured in **hours saved, not percentages**.

02 THE PLATFORM

One command centre, five AI pillars, one rule the AI can't break

SiteMind sits over a project's documents, specs, schedule, procurement, and test records as a single command centre. Five AI pillars do the work; a connective layer ties their findings into one living view, so an issue caught in compliance shows up again in the copilot, the schedule, the supply chain, and the timeline.



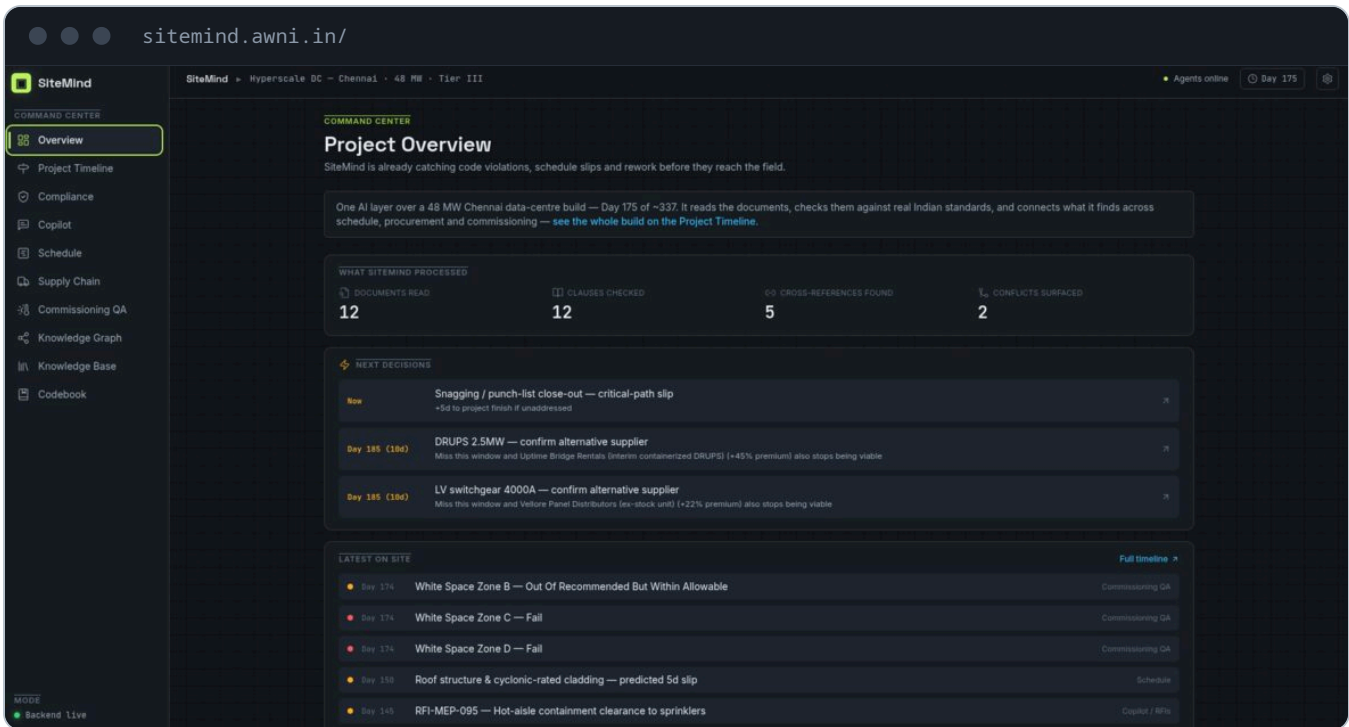
Site & design engineer

Catches the deviation and cites the clause — **Compliance** and the **Copilot**. Reads a submittal, gets the failing parameter with its source sentence, asks a cross-document question.



Project manager

Sees the blast radius of that one issue — the **Timeline**, **Schedule**, and **Supply Chain**. One deviation becomes a re-forecast finish date and an at-risk shipment.



The Command Center. One view over a 48 MW Chennai build: every pillar in the left rail, what the platform has already processed up top, the decisions due next, and a live feed where a finding from any pillar surfaces. The status reads **Backend live**, not mock data.

THE FIVE PILLARS

- ◆ **Compliance Agent** reads a submittal, extracts each parameter with its source sentence, and flags non-conformances against the exact cited Indian-standard clause.
- ◆ **Project & RFI Copilot** answers technical and contractual questions across every project document with citations, and spots when a similar RFI was already resolved.
- ◆ **Schedule Risk** forecasts the critical path, raises early warnings weeks ahead, and runs three specialist agents that propose real mitigation options.
- ◆ **Supply Chain Visibility** tracks critical shipments across suppliers, propagates delays, and alerts on at-risk deliveries with the lead time it computes for each.
- ◆ **Commissioning QA** turns real cooling test logs into pass, allowable, or fail verdicts against a thermal envelope, and builds the as-commissioned quality package.

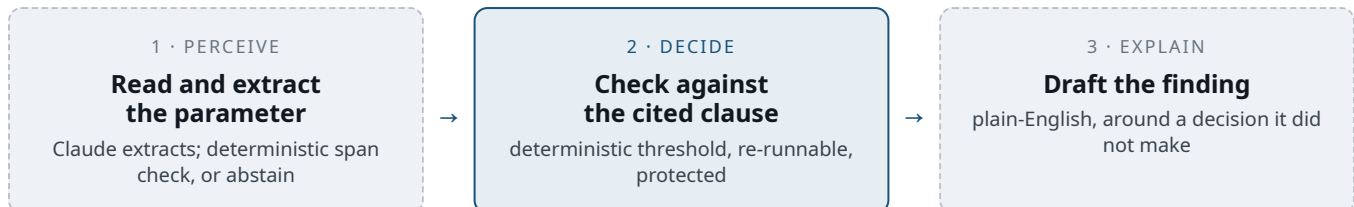
THE CONNECTIVE LAYER & THE STANDARDS BACKBONE

- ◆ **Command Center & Project Timeline** aggregate every finding from the five pillars into one ranked, cross-linked view, adding no new judgement of their own.
- ◆ **Knowledge Graph** is a deterministic **NetworkX** graph that traces every parameter to the physical asset, the RFI, and the schedule task it touches, a clickable memory rather than an unstable vector guess.
- ◆ **Codebook** is a standalone, **MCP-consumable** standards service, its own FastMCP server, that any agent (not only SiteMind) can query for real digitised clauses. This is the digitised-standards backbone.

- ◆ **Knowledge Base** is a multi-corpus **hybrid-retrieval** engine (BM25 keyword fused with dense embeddings) with live document upload, so a company's own submittals become searchable under the same cited, abstaining discipline.

THE DESIGN PRINCIPLE: PERCEIVE, DECIDE, EXPLAIN

Most AI compliance tools do one of two things. They hallucinate a plausible-looking citation, or they hide the judgement inside a model that just says "trust me." SiteMind splits the pipeline into three stages and lets a model touch only the outer two. The middle stage, the actual judgement, stays deterministic:



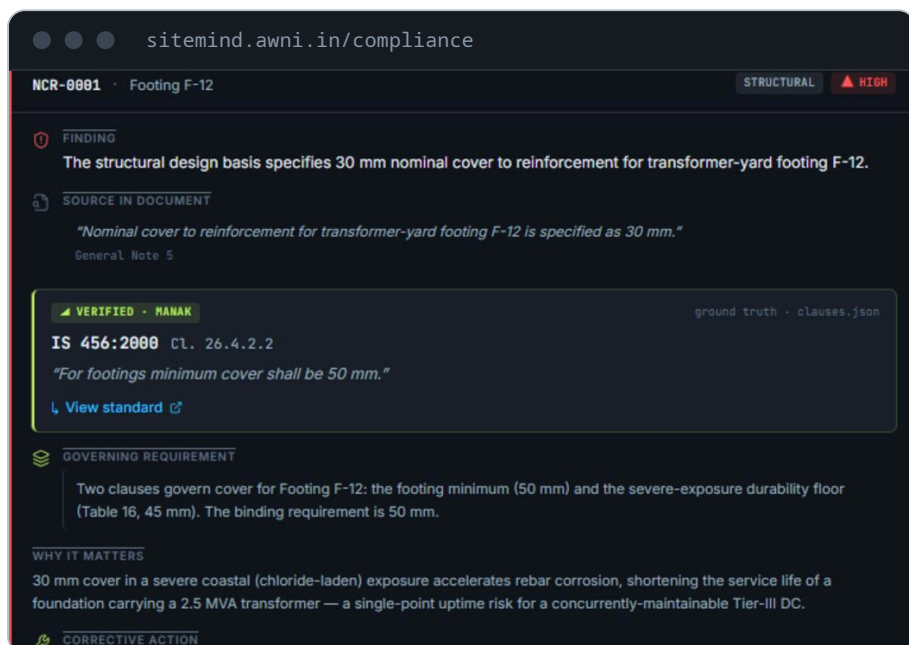
- 1 **Perceive** is where intelligence belongs. Claude (Sonnet 4.6, via the Claude Agent SDK) reads the submittal and pulls each value *with the exact sentence it came from*. A deterministic gate then confirms that sentence really appears in the document and the value sits inside it, or the reader abstains. So it can read a document it has never seen, yet is never allowed to invent a number.
- 2 **Decide** never changes. A plain Python threshold compares that value against the cited clause. The same input always gives the same verdict, so a judge can re-run any call by hand. This is the one stage no model is allowed to touch.
- 3 **Explain** hands the result back to Claude, which writes the finding in readable English, *around* a decision it was never permitted to make.

"The model perceives and explains. It never decides. Pass or fail is deterministic Python against a real, cited Indian-standard clause."

- ✓ **Never invent a clause.** Every citation resolves to real primary-source text: digitised IS and CEA codes, or verbatim from real BIS PDFs. Nothing is paraphrased from model memory.
- ✓ **Every clause shows its own reliability** through a source tag (`codebook_verified`, `primary_native_pdf`, `primary_scan_ocr`, `cross_source_unverified`), so trust tiers are shown, never quietly equated.
- ✓ **When unsure, it abstains.** Anything not confidently extracted is left out rather than guessed. The measured rate of fabricated values is zero.

What each pillar does, what's real, and what it scored

Each result below is a computed number from a named eval script. "What's real" is stated plainly, so there is no confusion about which parts are engineering and which parts are demo data.



The thesis, on screen. One non-conformance from the Compliance hero: the model pulled **30 mm cover** with its **source sentence**; deterministic Python then failed it against the **real cited clause** — IS 456:2000 Cl. 26.4.2.2, "minimum cover shall be 50 mm." Perceive, decide, explain, in a single card.

PILLAR	HOW IT WORKS, AND WHAT IS REAL	RESULT
1 • Compliance Agent HERO agents/compliance.py	Reads a design basis or submittal, extracts each parameter with its source span, and checks it against a real cited IS or CEA clause. Emits non-conformances plus a senior action brief, across structural and electrical domains. Real: LLM extraction (Claude Sonnet 4.6) behind a span-verification gate, the 24 digitised clauses, the deterministic checks, and the abstention floor.	100% acc 0% halluc.
2 • Project & RFI Copilot agents/copilot.py	Guardrailed hybrid retrieval, BM25 keyword fused with dense embeddings by reciprocal-rank fusion , across every project document, plus detection of RFIs already answered before. A deterministic cosine floor is the go/no-go gate: below it the copilot abstains instead of guessing. Real: the fusion, the eval-calibrated floors, and the abstention behaviour.	100% n=21
3 • Schedule Risk schedule.py · mitigation.py	Critical-path analysis plus early-warning rules (slipping vendor, progress lag, monsoon window, festival workforce dip), then a recomputed finish date and three specialist mitigation agents (procurement, float, resource). Real: the CPM maths and every rule. No fabricated-data ML.	100% mitig. 13/13
4 • Supply Chain supply_chain.py	Multi-tier shipment tracking, delay propagation, root-cause attribution, severity-tiered alerts with real lead time, and an equipment-spec check (IS 8623-1 switchgear). Real: the propagation, root-cause, and alternative-viability maths.	100% alerts 11/11

PILLAR	HOW IT WORKS, AND WHAT IS REAL	RESULT
5 • Commissioning QA commissioning.py	Real cooling test-log CSV into pass, within-allowable, or fail verdicts against a thermal envelope, then an exportable quality package. Real: the per-row threshold logic. Cooling is built; electrical and fire are deferred on a disclosed corpus gap.	100% n=14

THE STANDARDS BACKBONE & CONNECTIVE LAYER

CODEBOOK MCP

A standalone standards service with its own MCP server
(`list_corpora`,
`search_standards`,
`get_clause`), so any external agent can query the same digitised backbone. Verified through a live client session.

KNOWLEDGE BASE

Multi-corpus hybrid retrieval (BM25 + dense embeddings, reciprocal-rank fused) with live document upload, so a company's own submittals become searchable under the same cited, abstaining discipline.

KNOWLEDGE GRAPH

A deterministic NetworkX graph linking equipment, spec, standard, and RFI, so the cross-references between pillars are traceable and clickable, not a vector guess. Project Timeline aggregates every finding over it.

04 THE EVIDENCE

Every correctness claim is a number we can re-run

21 re-runnable eval scripts, 18 in the backend and 3 in the Codebook service. Each writes a report with its case counts and accuracy. None are blended into a single vanity score; every pillar is reported on its own.

The flagship result: compliance decision accuracy

On 41 hand-built boundary cases (including tight `>=`-versus-`>` edges and not-applicable cases), SiteMind's grounded checks are compared against a naive keyword baseline, with gold labels taken from the real IS-clause thresholds.



Macro-F1 **1.00 versus 0.56**. The confusion matrices tell the story more plainly than any single number: SiteMind lands entirely on the diagonal, while the baseline leaks errors into every off-diagonal cell. Citation-hallucination rate **0.0%**: all 32 clauses cited in the run were independently re-resolved to their real IS text.

SITEMIND				NAIVE KEYWORD BASELINE			
41 / 41 correct – pure diagonal				24 / 41 correct – errors everywhere			
	V	C	NA		V	C	NA
V	16	0	0	V	12	1	3
C	0	16	0	C	4	8	4
NA	0	0	9	NA	4	1	4

Correct (on the diagonal)
Misclassified
rows = actual · cols = predicted · V·iolation / C·ompliant / N·A

These are **deterministic checks, not machine-learning predictions**, so a perfect score means the arithmetic is exactly right. The numbers that carry weight are the jump over the baseline (58.5% to 100%) and the zero-hallucination rate, both of which come from real comparison, not self-report.

Extraction: the no-fabrication guarantee, measured

14 held-out mini-documents with unseen phrasing, including adversarial decoys where a trigger word appears with no valid value and must not be extracted:

1.00
 Precision, recall, and F1 on 11 planted parameters

0
 Fabricated values, the metric that matters most

59/59
 Correct abstentions, never a false value

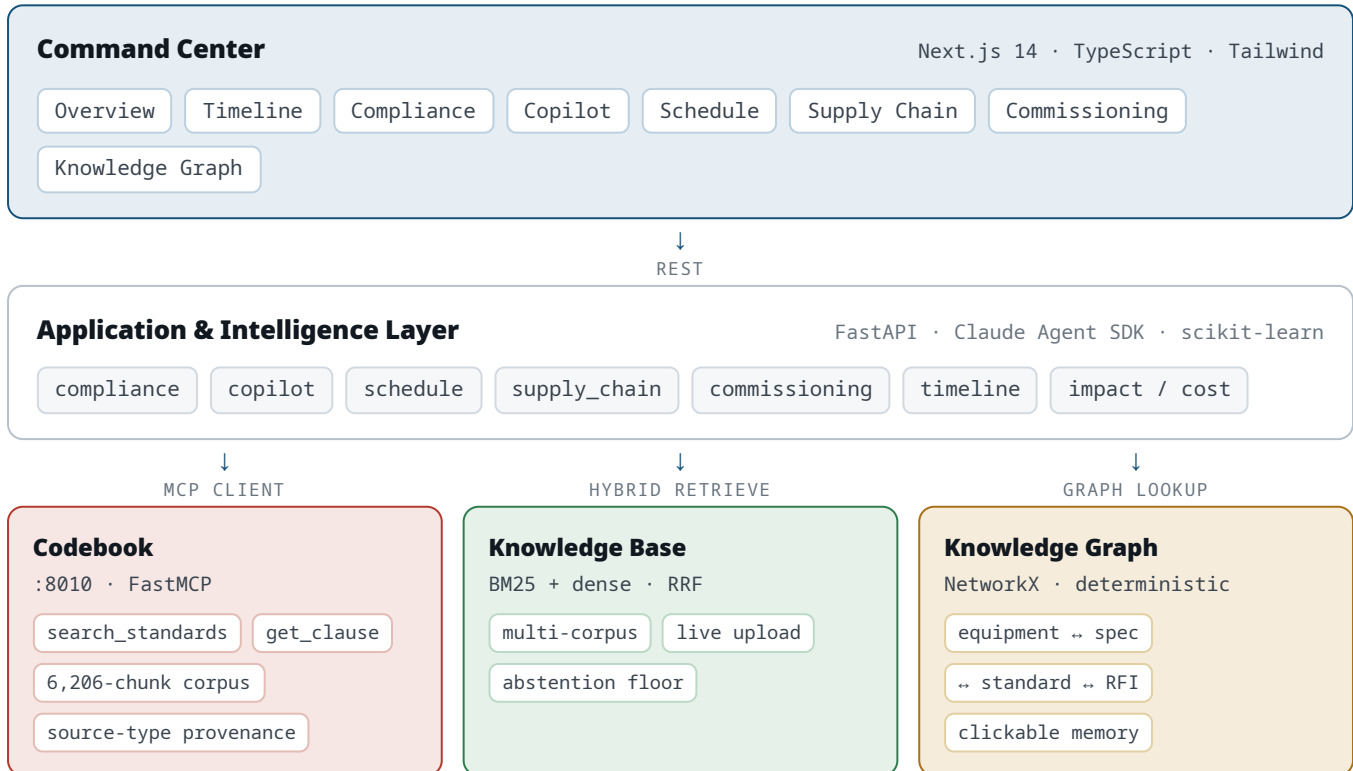
The full suite at a glance

eval area	what it proves	cases	result
Compliance rules	Structural rule engine + citation grounding vs baseline	41	100%
Extraction	Parameter extraction, zero fabrication, correct abstention	14 / 59	F1 1.0
Electrical	Electrical checks against IS 732 thresholds	32	100%
Equipment spec	IS 8623-1 switchgear spec matching	12	100%
Commissioning	Cooling-envelope pass, allowable, and fail verdicts	14	100%
Supply chain + alerts	Delay propagation, root cause, alert severity, lead time	8 / 11	100%
Schedule + factors	CPM re-computation, monsoon and workforce arithmetic	33	100%
Mitigation agents	Three agents, verdict and days-recovered	13	100%
Timeline	Cross-pillar traceability on the live dataset	20	100%
Cost + impact	Cost-at-risk and ROI-ticker composition	21	100%
Copilot + retrieval	Floor calibration, RRF fusion, end-to-end ingest and query	50	100%
Corpus integrity	6,206 real chunks byte-verified verbatim, read-only proof	26	100%
Codebook (MCP, live)	All 4 standards tools through a live client session	30	100%

The strongest scripts test against real derived or external data with full coverage, not hand-picked cases: the timeline eval runs on the live project, and the corpus eval byte-checks every one of the 6,206 chunks. No script backtests a prediction against real historical outcomes, because no such project history exists, and the schedule, supply-chain, and cost evals say so plainly.

Deliberately simple, deliberately auditable

A command centre in the browser talks to a Python API. Each pillar is a bounded service over real structured data, and the standards backbone is a separate service the backend consumes as a client.

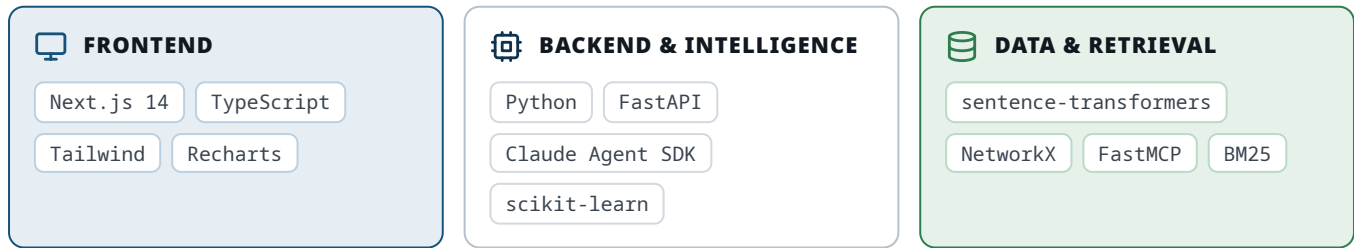


RETRIEVAL AND THE STANDARDS BACKBONE

The intelligence layer never talks to a monolithic "vector database." It fans out to three purpose-built backbones, each auditable on its own:

- ◆ **Codebook** is a standalone service that runs its own **FastMCP** server over streamable-HTTP, exposing `list_corpora`, `search_standards`, and `get_clause`. The SiteMind backend consumes it as a client, and so can any other agent in an EPC stack, so the verified-standards layer is shareable, not locked inside one app.
- ◆ **The Knowledge Base and the Copilot** both run the same **guardrailed hybrid retrieval**: BM25 keyword ranking fused with dense embeddings by **reciprocal-rank fusion**, with a deterministic dense-cosine floor as the abstention gate. Fusion decides *which* passages surface; the floor decides whether there is enough to answer at all, and that go/no-go stays deterministic, in keeping with the rule that the model never decides.
- ◆ **The Knowledge Graph** is a deterministic NetworkX graph, not a vector store: it traces a parameter to the physical asset, the RFI, and the schedule task it affects, so a cross-pillar link is a fact you can follow, never a similarity guess.

BUILT WITH



SiteMind is **LLM-first where intelligence matters, and deterministic where trust matters**. Two layers, one clear boundary between them:

- ◆ **The intelligence layer (Claude Sonnet 4.6, via the Claude Agent SDK):** reads and extracts from documents, answers cross-document questions, weighs trade-offs, and writes every finding and answer in plain language. This is where the product feels smart.
- ◆ **The guarantee layer (deterministic Python):** issues the compliance verdict and resolves every citation to real clause text. This is the one place a model never acts alone, so a verdict or a citation can never be fabricated.
- ◆ **Runs where the data does:** the guarantee layer is offline-capable, so SiteMind still stands up in secure, air-gapped data-centre environments. The intelligence layer runs when a provider is configured, with a deterministic fallback so nothing ever breaks without it.

AGENTIC WHERE THE WORK IS OPEN-ENDED, VERIFIED WHERE IT IS A VERDICT

SiteMind uses LLM agents wherever a task is genuinely open-ended: reading a novel submittal, answering a cross-document question, or weighing several valid mitigation options. The brief names multi-agent specifically for generating mitigation options, and that is exactly where SiteMind is agentic: three specialist agents (procurement, float, resource) each make one grounded computation, and a coordinator collects every result, including the non-viable ones.

The one hard line is the graded verdict. A single pass-or-fail against a cited standard stays a deterministic check, so an agent can never drift into a plausible-but-wrong answer. Everything the model produces is **grounded**: its extractions against a source span, its answers against retrieved text, its verdicts against a deterministic check.

06 IMPACT & SCALABILITY

Value measured in hours, breadth added by clauses

BUSINESS IMPACT

The Overview computes return per pillar from the platform's own outputs: issues caught, engineer-hours saved, and rework value avoided, with every term of the cost-at-risk formula shown rather than a single headline.

On this 48 MW build, that composes to about 174 engineer-hours recovered and roughly ₹1.06 crore of rework exposure avoided — six compliance non-

SCALABILITY

The compliance engine is clause-driven: every check maps to a real digitised clause, so **breadth grows by adding a clause, never by retraining a model**. There is no training run to reproduce, no dataset to relabel, and no drift to monitor.

- ◆ The standards service already holds **6,206 verified chunks** across 17 real Indian codes, ready for new

conformances plus the schedule, supply-chain, and commissioning findings, each priced by its own disclosed per-issue constant and summed live on the Overview. This answers the brief in its own terms: **hours, not percentages**.

rules to point at.

- ◆ It is exposed over **MCP**, so any external agent, not just SiteMind, can consume the same verified standards backbone.
- ◆ Next codes in line: IS 875 wind loads, IS 13920 seismic detailing, IS 800 steel, and NBC 2016 fire and egress.

07 INTEGRITY & LIMITS

What's real, what's representative, what's not done yet

Disclosing this is the point, not a footnote. A tool built on "the model never decides" has to hold itself to the same standard, so here is the honest boundary of the prototype.

REAL

- ◆ Every IS and CEA clause citation, and the text it resolves to.
- ◆ The compliance decision logic and all 21 evals.
- ◆ Document ingestion: an uploaded PDF or DOCX is genuinely parsed, with mandatory abstention.
- ◆ The critical-path re-computation and every pillar's arithmetic.
- ◆ The 6,206-chunk standards corpus, byte-verified word for word.

REPRESENTATIVE

- ◆ The pre-loaded project documents and schedule are synthetic, modelled on real public Indian data-centre tenders.
- ◆ The standards and the logic that checks them are real, and so is anything you upload live.

KNOWN LIMITS

- ◆ Some clause verify-links point at a dev host; the clause text still resolves in-app, so not every link is publicly clickable yet.
- ◆ Commissioning is cooling-only; electrical and fire wait on a corpus gap (NBC 2016, DG-set testing).
- ◆ ROI constants (about 20 engineer-hours and about ₹15 lakh per issue) are labelled assumptions, not measurements.

08 WHY IT STANDS OUT

Five reasons SiteMind is different from a demo

IT CAN'T HALLUCINATE A CITATION

Every clause is real primary-source text with a disclosed reliability tier. The measured hallucination rate is zero, not a hope.

THE AI NEVER MAKES THE CALL

Pass or fail is deterministic Python against a cited clause, so the credibility-critical step is testable and reproducible, offline.

EVERYTHING IS A REAL NUMBER

21 re-runnable evals, a 100% versus 58.5% comparison, and 6,206 byte-verified clauses. Nothing on a slide is asserted.

IT SCALES WITHOUT RETRAINING

Add a clause to widen coverage. No model to retrain, no dataset to relabel, and a standards backbone any agent can query over MCP.

IT IS HONEST ON PURPOSE

Synthetic project data and assumption-based ROI are disclosed openly, not hidden. For zero-tolerance infrastructure, a tool you can audit beats a tool that sounds confident.

09 THE ROAD AHEAD

From a working prototype to the layer under the buildout

NEAR TERM

- ◆ **Extend the perception layer to drawings:** the text path already runs Claude behind the span-verification gate; add vision models so shop drawings and stamped submittals read the same way.
- ◆ Widen the code library to IS 875, IS 13920, IS 800, and NBC 2016, the most-cited code in real permitting.
- ◆ Extend Commissioning QA from cooling to electrical and fire once those standards are digitised.
- ◆ Publish public verify-links for every clause, closing the last known gap.

THE BIGGER PICTURE

Because the standards backbone speaks MCP, SiteMind is not one closed app. It can become the verified-standards layer that any tool in an EPC stack calls, from a procurement system checking a submittal to a commissioning app validating a test log.

THE CLOSE

India is building over 2,700 MW of data-centre capacity by 2027, at zero tolerance for the errors that quietly compromise uptime. **SiteMind doesn't ask anyone to trust an AI. It shows its work, cites its source, and lets you check every call it makes.** That is what makes it ready for the infrastructure the country is actually building.